

Highlights

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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- Voxel coupling averaged over the source cell equals the continuum's exactly
- The cube's single-site T-matrix follows from distributional moment integrals
- The voxel scheme's leading error is known in closed form for every grid
- A first-moment voxel converges at fourth order for P, SV and SH waves
- In a random stratified layer, one voxel per model layer reaches 1e-6 accuracy

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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Abstract

We analyse the convergence of voxel discretisations of the elastic volume integral equation, in which a medium is divided into cubic cells, each represented by a single-site T -matrix and coupled through the background Green's tensor: the elastic counterpart of the discrete dipole approximation. Because cubes tile space, a layer of identical voxels is exactly a homogeneous layer, so it isolates the discretisation error, free of modelling and shape error. We show that (i) the cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling equals the continuum's and the lattice kernel at normal incidence is in closed form; (ii) the single-site T -matrix follows from distributional moments of the Green's tensor and, against exact scattering by a sphere, is exact statically and departs only by a real, closed-form $(ka)^2$ term; (iii) its self term cancels, so the scheme is a collocation of the continuum equation, of second order with leading error $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission, which no scalar correction can remove; (iv) a Galerkin voxel carrying the first moment of its internal field converges at fourth order, with leading error $-(kd)^4/720$, for incident P, SV and SH

waves at normal and oblique incidence, and its spectral lattice sum needs no Ewald splitting. In a randomly stratified layer the orders are unchanged, and the fourth-order voxel reaches an accuracy of 10^{-6} with one voxel per model layer. Every result is derived symbolically and verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves.. Seismic waves in the crust are scattered by heterogeneity on every scale. The models that describe it are parameterised in many ways — as layers or blocks, by smooth basis functions, or by interfaces — and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation represents the sampled model as cells of constant properties and treats every cell as a scatterer: a voxel carrying a single-site T -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green’s tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the

17 single voxel's response is matched to cancel it, refinement converges to the
18 wrong answer without revealing it.

19 *Why a homogeneous layer..* This paper therefore studies the one configura-
20 tion in which a voxel scheme runs in full — every voxel, every coupling, the
21 whole multiple-scattering series, at normal and at oblique incidence — and
22 the exact answer is nevertheless known: a laterally uniform layer filled with
23 identical voxels. Because cubes tile space, such a layer does not approximate
24 a homogeneous layer; it *is* one. Any difference between its reflection and
25 transmission and those of the continuous layer is therefore error of the dis-
26 cretisation alone, with no modelling error and no shape error behind which
27 to hide. A sphere, the usual test object for volume-integral and discrete-
28 dipole methods [1], cannot give this: its error mixes the discretisation with
29 the staircase approximation of a curved surface by cubes. The layer plays the
30 part of the patch test in finite elements. A scheme that cannot reproduce a
31 uniform medium correctly cannot be trusted in a heterogeneous one. What
32 the layer reveals about the scheme's two ingredients — the coupling between
33 voxels and the response of a single voxel — does not depend on the medium,
34 because the same two ingredients are used wherever the scheme is applied.

35 *What the reader gains..* Four things that a heterogeneous test cannot pro-
36 vide. First, a proof, exact at normal incidence, that the voxel scheme con-
37 verges to the continuum, and the specific ingredient — a coupling averaged
38 over the source voxel — without which it converges to the wrong answer. Sec-
39 ond, the leading error in closed form for every discretisation, so that a grid
40 can be chosen for a required accuracy before the calculation is run. Third,
41 a single-site T -matrix derived from first principles, with its error stated and

42 checked against the exact solution for a sphere. Fourth, a voxel that converges
43 at fourth order, also in a randomly stratified model (§9): for a reflection ac-
44 curacy of 10^{-6} it allows voxels 58 times larger in each direction than the
45 standard one (§10).

46 *Why cubic voxels..* Cubes are the natural cell for three reasons. They *tile*
47 *space*, so a uniform region discretised into cubes is still that uniform region,
48 whereas spheres or point scatterers can only approximate it and carry a pack-
49 ing error of their own. Their *regular lattice* makes the interaction between
50 cells translation invariant, so the coupling is a convolution — Toeplitz be-
51 tween planes, Bloch-periodic within them — and the whole system is solved
52 by fast transforms. And a cube small compared with the wavelength has a
53 *single-site response in closed form*, the elastic depolarisation of a cube (§6),
54 so the only approximation a voxel introduces is the internal field it assumes.

55 *Multiple scattering on a lattice..* Replacing a heterogeneous body by a lattice
56 of small scatterers that interact through the background Green’s function
57 is the multiple-scattering description of Foldy [2] and Lax [3]. In electro-
58 magnetics its discretised form is the discrete dipole approximation [DDA;
59 4, 5], whose accuracy has been studied in detail: each dipole’s polarisability
60 must account for the local field of the surrounding lattice, which the lattice
61 dispersion relation fixes so that an infinite lattice propagates waves as the
62 continuum does to a given order in kd [6]; the error of the method converges
63 at second order in the dipole spacing when shape errors are absent [1]; and
64 integrating the Green’s tensor over each cell rather than sampling it at the
65 centre improves accuracy, most of all for strong contrasts [7]. The elastic
66 problem adds a structure the scalar and electromagnetic ones lack. A cubic

67 voxel responds to the incident field with a force (through its density con-
68 trast) and a moment (through its stiffness contrast), so the state at a voxel
69 has nine components — displacement and strain — and the single site’s self-
70 interaction is the elastic depolarisation of a cube, the discrete counterpart of
71 the inclusion tensor of Eshelby [8].

72 *This paper..* The approach originates in the two-and-a-half-dimensional for-
73 mulation of Nestor [9]: a medium divided into cells that interact through the
74 background Green’s tensor, each carrying a single-site contrast renormalised
75 by its own near field, with the multiple-scattering series summed. This pa-
76 per extends it to cubic voxels. §2 sets out the discrete medium and §3 the
77 continuous layer it must reproduce. §4 shows why the point coupling fails,
78 and §5 why the source-averaged coupling succeeds exactly. §6 derives the
79 single-site T -matrix from moment integrals, tests it against the exact sphere,
80 and shows that with it the discrete system is a collocation of the continuum
81 equation. §7 gives that collocation’s second-order error in closed form, and
82 §8 the fourth-order voxel, for incident P and S at normal and oblique inci-
83 dence. §9 repeats the tests on a randomly stratified layer, and §10 discusses
84 what the results imply for voxel schemes in general. Every result is derived
85 symbolically in Mathematica and checked against an independent Python
86 implementation or an exact solution.

87 2. Setting

88 *The discrete medium..* Space is tiled by cubes of side d (half-side $h = d/2$).
89 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
90 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus

91 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 92 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 93 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

94 displacement followed by engineering strain. A voxel responds with a force
 95 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 96 state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source
 97 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

98 with ψ^0 the incident state. Strain rows of $\mathbf{K}$ are receiver derivatives of the
 99 point Green's tensor $\mathbf{G}$; moment columns are derivatives with respect to the
 100 receiver coordinate, symmetrised with a factor $\frac{1}{2}$ on the shear components.
 101 In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (3)$$

102 where $\Delta \mathcal{C}$ is the 6×6 contrast stiffness with $\Delta \lambda + 2\Delta \mu$ and $\Delta \lambda$ in the normal
 103 block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

104 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 105 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 106 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 107 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 108 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (4)$$

the sum running over the square lattice $\mathbf{R}$ of pitch d , m the plane separation in pitches. The question is whether, and how fast, this chain converges to the continuous layer as $n \rightarrow \infty$ at fixed D .

Implementations.. Every result is derived symbolically in Mathematica and compared with an independent numerical implementation or an exact solution. “The package” below is the Python implementation of the voxel scheme (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified propagator, which serve both as the object under test and, where validated, as a second implementation.

3. The continuous layer and its thin-layer expansion

At normal incidence a plane force drives a one-dimensional problem per wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission follow from continuity of displacement and traction at its faces; they agree with the stratified propagator used as the reference in this work to 2.5×10^{-8} , P and S (`ContinuumLimit_Reference.wl`).

For a layer of thickness D with modulus M_1 and density ρ_1 in a background M_0, ρ_0 , the reflection coefficient expands as

$$R = \frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] + O(D^2), \quad (5)$$

exact in the contrast (`ContinuumLimit_ThinLayer.wl`): to first order a thin layer is an interface across which the traction jumps by the excess inertia and the displacement by the excess compliance. At oblique incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$ for the P–SV state

131 $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$,
 132 which reproduces the layer's $O(D)$ reflection coefficients R_{PP}, R_{PS} to 10^{-12} ;
 133 the series truncated at $O(D^k)$ converges at exactly order k ; and a soft thin
 134 layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η tends to the linear-
 135 slip interface as $D \rightarrow 0$. These expansions are the target a single plane of
 136 voxels must reproduce order by order.

137 4. The specular sums of the point Green's tensor

138 By Poisson summation,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (6)$$

139 with $\hat{\mathbf{G}}$ the two-dimensional transform of the Green's tensor between planes.
 140 The $\mathbf{g} = 0$ term is the continuum's plane-wave term; the others are evanes-
 141 cent, decaying as $e^{-2\pi m \sqrt{p^2 + q^2}}$ independently of d . In the strain-from-moment
 142 block they carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel vol-
 143 ume does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 144 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 145 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 146 planes in slab order, the static lattice sum of the point kernel is exactly the
 147 depolarisation of a uniform plate — $-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 148 on the normal shears, per unit volume — plus a cubic-symmetric remainder
 149 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel there-
 150 fore reproduces the continuum only if that remainder is cancelled by the
 151 voxel's own self term.

152 5. The tiling identity

153 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$
 154 $\mathbf{y}) d\mathbf{y}$ — the field at a voxel centre of a uniform polarisation of the source
 155 voxel. Each Poisson order in Eq. (6) is then multiplied by the cube’s form
 156 factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (7)$$

157 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 158 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 159 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube’s indicator χ . Two exact statements follow.

160 *Between planes..* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (8)$$

161 the continuum’s plane-wave term times the vertical form factor. The pack-
 162 age’s cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 163 `AveragedSums.wl`).

164 *Within a plane..* The sum over *all* cells of the plane, the voxel’s own included,
 165 is the field at the mid-plane of a uniform plate of thickness d :

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (9)$$

166 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 167 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 168 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 169 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
 170 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for

171 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 172 with every cell integral evaluated analytically (below).

173 *Consequence: an exact kernel with no lattice sum..* At normal incidence the
 174 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (10)$$

175 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 176 — Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 177 Taylor tail beyond it — carries a static strain-from-moment error of 9.4×10^{-4}
 178 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 179 and the quadrature order are raised; Eq. (10) is now the package’s kernel at
 180 normal incidence.

181 *Cell integrals without quadrature..* Write the static (Kelvin) tensor as $G_{bc} =$
 182 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 183 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 184 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,
 185 which never contain the origin, and there the two-dimensional antiderivative
 186 in the face coordinates is continuous, so the face integral is a sum over the
 187 face’s corners. The distributional term at $r = 0$ is carried by the divergence
 188 theorem itself. (A triple antiderivative evaluated at the cube’s corners is
 189 *not* usable for repeated derivatives: its arctangent terms jump across the
 190 coordinate planes through the origin, and differentiating twice there discards
 191 exactly that term.) The self term so obtained reproduces the classical closed
 192 form of the cube’s depolarisation to machine precision.

193 6. The single site: the cube T -matrix from moment integrals

194 Every error of a voxel scheme is made at the single site (§10), so the single
195 site is derived here rather than imported. The classical route imports it. For
196 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a
197 uniform load uniform, with a depolarisation tensor in closed form [8, 10, 11];
198 its dynamic generalisation is known for spheres and ellipsoids [12, 13]. For
199 a cube or a general polyhedron the static field of a uniform eigenstrain is
200 also in closed form [14, 15, 16, 17], but the strain it produces is not uniform,
201 so there is no exact uniform-strain inclusion to import. In scattering theory
202 the same object appears as the self-consistent or effective-field closure of
203 the volume integral equation [18, 19, 20]: a first-order (Born) scatterer [21]
204 renormalised by the geometric series of its own near field, the step that turns
205 a contrast into an effective contrast. Nestor [9] introduced the single site
206 in this form, summing the series of the cell’s self integral, with the singular
207 near field evaluated over an infinitesimal cell. The moments below evaluate
208 the same self integral over the finite cube, as a distribution. The elastic T -
209 matrix of Waterman [22] represents a scatterer in spherical multipoles, the
210 natural basis for a sphere; a voxel in a lattice needs one compatible with the
211 lattice, and the moments over the cube provide it. The construction below
212 makes the approximation of a voxel explicit rather than implicit: it derives
213 the cube’s T -matrix from the moments of the elastodynamic Green’s tensor
214 over the cube, and identifies the uniform-strain closure as one truncation of
215 an exact hierarchy.

216 *6.1. The closed hierarchy*

217 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 218 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second
 219 degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q .
 220 The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed
 221 system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (11)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (12)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (13)$$

222 all fields evaluated at the centre. Its coefficients are six moments of the
 223 background Green’s tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \end{aligned} \quad (14)$$

224 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
 225 reaches $\mathbf{N}$ and $\mathbf{Q}$ only, and no symmetry, parity or scaling check can detect
 226 it, so it is fixed by convention. The moments are graded by (D, W) , the
 227 number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$,
 228 $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$.

229 *6.2. The moments as distributions*

230 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (15)$$

231 [23, 24]. Expanding g_c in powers of r reduces every moment in Eq. (14) to a
232 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (16)$$

233 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
234 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
235 gral depends on the shape of the region excluded around the origin. This is
236 the source of the δ -function term of the inclusion problem, which a numerical
237 quadrature misses. Without it the depolarisation is wrong; for a sphere it
238 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
239 are distributional from the start, and each is moved onto the voxel's faces by
240 pairing with the test function $w\mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w\mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (17)$$

241 The surface term is evaluated on the faces, a distance h from the origin, where
242 every derivative of r^m is smooth. The recursion lowers D and W together
243 and ends on convergent integrals. The face integrals are done in closed form
244 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (18)$$

245 which the face route satisfies exactly and a spherical excision returns as
246 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell

247 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
 248 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
 249 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
 250 Because every power of r is paired this way, the dynamic moments keep the
 251 even-power terms of the expansion (k^2r , r^3 , ...) as well as the odd ones.
 252 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
 253 those terms, at a real $O((kh)^2)$ cost (§6.4).

254 6.3. Parity, and the nine-component block

255 The cube is centrosymmetric. A moment that couples a field quantity of
 256 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 257 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (12) contains the first gradient alone: ∂u is
 258 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 259 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 260 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 261 by Eq. (12), and *no treatment of the second-gradient block (13) can change*
 262 *it*. What Eq. (13) contributes reaches the far field only through the $O((kh)^2)$
 263 coupling in Eq. (11). That is the order, and the origin (the variation of the
 264 field across the voxel), of the collocation error of §7, which the first-moment
 265 voxel of §8 removes.

266 With $\Delta\rho = 0$, Eq. (12) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk} \delta c_{nkrj}$, built
 267 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 268 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are

269 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (19)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (20)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (21)$$

270 The density channel is the displacement block of Eq. (11) with the second
 271 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 272 of $\mathbf{G}$.

273 *The cube against the sphere..* For a sphere, Eshelby's tensor gives a dilata-
 274 tional channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 275 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 276 latational channel, Eq. (19), is the sphere's exactly: its eigenvalue involves
 277 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 278 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 279 the shape. The cube's shear channels differ from each other. Their isotropic
 280 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (22)$$

281 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 282 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
 283 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
 284 and the diagonal shear is the more depolarised. At the background of §7

285 $(\nu = 7/32)$, $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
 286 0.4960.

287 *Where Eshelby enters..* Equations (19)–(21) follow from the moment inte-
 288 grals and the group theory alone. No ellipsoidal result is used, and no δ term
 289 is added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 290 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
 291 the uniformity theorem that ansatz is exact if and only if the inclusion is
 292 an ellipsoid. For a sphere the construction is therefore exact in the static
 293 limit, and for a cube it is an approximation. The hierarchy (11)–(13) states
 294 that approximation as the omitted blocks, rather than as an unquantified
 295 shape correction. A polynomial expansion about the centre cannot represent
 296 the edge and corner concentrations of an isolated cube's internal field at any
 297 finite degree. A voxel in a uniform layer has no such concentrations: its
 298 neighbours have the same properties, so its internal field is the continuum's,
 299 which is smooth. That is why a low-degree internal field suffices in a lattice
 300 (§§7–8) although it is only an approximation for an isolated cube.

301 *Against the exact sphere..* These statements are tested against the exact
 302 elastic Mie solution for a sphere, with no Eshelby formula in between. The
 303 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
 304 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
 305 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
 306 route of Eq. (17) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
 307 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
 308 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 309 isotropic shear average, Eq. (22), is the depolarisation read from Mie's E_2

310 to 10^{-8} . The same moment taken as a principal value, over the ball with
 311 a small sphere excluded around the origin, has zero depolarisation in every
 312 channel. The whole static response of the sphere is the δ term.

313 At finite frequency the closure uses the dynamic ball moments, which are
 314 in closed form. It departs from Mie at second order in ka and first order in
 315 the contrast, and the departure is closed-form too: the ball's moments are
 316 elementary, and Mie's low-frequency series with the contrast kept exact gives
 317 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta \lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta \rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta \mu}{\mu}, & n = 2, \end{cases} \quad (23)$$

318 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 319 the radiation damping, agree exactly in all three channels. So the closure's
 320 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 321 own. That is the order the hierarchy assigns to the omitted second gradi-
 322 ents (§6.3), and it has the same origin, the variation of the field across the
 323 scatterer, as the collocation error of §7. Eq. (23) is derived symbolically
 324 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 325 the numerical Mie solve and an independent implementation of the closure
 326 to 5×10^{-4} , which is the size of the next order in the contrast.

327 6.4. The collocation closure

328 The collocation closure takes the internal field of a voxel as uniform, the
 329 truncation of §6.3 for the density and stiffness channels together. The voxel's

own field at its centre is then $\mathbf{S}_{\text{self}}\Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 So $e = \psi + \mathbf{S}_{\text{self}}\Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)^{-1}. \quad (24)$$

With the kernel (10) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (25)$$

an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
exact cell integral of the one-dimensional continuum kernel (Eqs. (8), (9)), so
 Eq. (25) is precisely the collocation of the continuum Lippmann–Schwinger
 equation with a strain piecewise constant over voxels. It does not contain
 $\mathbf{S}_{\text{self}}$ at all.

The package’s cube T -matrix is this closure statically: both blocks agree
 with Eq. (24) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$
 through its far-field form-factor corrections. Its self integrals are exact in the
 static and radiation parts but omit the real even-power terms of the Green’s
 tensor’s expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 self term, whose series agrees with the closed-form elastodynamic tensor to
 10^{-14} . By Eq. (25) that error cannot enter the closure scheme.

348 7. Convergence, and the closed-form error

349 *Second order..* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
 350 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
 351 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
 352 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
 353 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (25) and with the
 354 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
 355 $k_S h$ alone.

356 *The leading error in closed form..* In Eq. (25) the kernel is integrated exactly;
 357 the only approximation is the internal field's variation across a voxel. At first
 358 order in the contrast the discrete scattered field is therefore a midpoint rule
 359 applied to the continuum's Born integral, with the kernel's cell integral exact.
 360 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 361 it is flat. With k the background wavenumber of the wave concerned (k_P for
 362 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 363 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (26)$$

364 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 365 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 366 and those of Eq. (29) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 367 At the full contrast the reflection constant is unchanged (0.124 against $1/8$);
 368 the transmission constant grows to -0.092 .

369 *Why a uniform-field single site stops at second order..* The two constants
 370 have opposite signs, so no scalar correction of $\mathbf{T}$ — no single $O((kh)^2)$ fac-
 371 tor — can cancel both. The package T -matrix illustrates it: its form-factor
 372 corrections reduce the reflection error by a factor 4.45 and *increase* the trans-
 373 mission error by 1.7, at every n . The missing term in Eq. (26) is the first
 374 moment of the internal field over each voxel, which radiates through the
 375 derivative of the kernel. A fourth-order scheme must carry it: a single site
 376 with a gradient channel, and a kernel whose cell integrals include the kernel's
 377 derivative.

378 8. Fourth order: the first-moment voxel

379 Let each voxel carry the mean *and* the first moment of its internal field.
 380 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_0 = 1$,
 381 $\phi_1 = s/h$, the field is $w = \sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and
 382 strain of the P channel, and the polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with
 383 $\Delta M = \Delta \lambda + 2 \Delta \mu$. Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (27)$$

384 with the same two functions on every voxel gives

$$\left(\int_i \phi_a^2 \right) c_{a,i} = \int_i \phi_a w^0 + \sum_{j,b} \left[\int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (28)$$

385 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
 386 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$ and
 387 the local term contributes $-\int_i \phi_a \phi_b / M_P$. Every entry is closed-form. Keeping

only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 converges at fourth order in both reflection and transmission (`ContinuumLimit_FourthOrder.w1`; Table 1). A single plane of first-moment voxels represents the 2 m layer

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (28). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

6500 times more accurately than a single plane of uniform ones. The collocation column is the scheme of §6 and reproduces the chain of the full 9×9 system to all printed digits.

The leading error in closed form.. Without scattering, the Galerkin solution is the L^2 projection Π of the incident field onto each voxel's basis, and the kernel is integrated over each voxel exactly; at first order in the contrast the error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} , $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] \, ds}{\int_{-h}^h e^{i(q+k)s} \, ds}, \quad (29)$$

401 with Π the centre value for collocation, the cell mean for the mean-only voxel
 402 and the projection onto $\{1, s/h\}$ for the first-moment voxel (`ContinuumLimit_`
`FourthOrderError.wl`). The results are in Table 2. The chain of Eq. (28)

Table 2: The leading error of the discrete layer in closed form, Eq. (29): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$

403

404 reproduces all six forms at $n = 1$ to 8 to 10^{-4} of the error, the remainder
 405 being the second order in the contrast. The first-moment voxel's leading
 406 error is the same in reflection and in transmission: its projection error is or-
 407 thogonal to everything linear across the voxel, so the leading term no longer
 408 depends on the direction in which the field is observed.

409 *In three dimensions..* The three-dimensional first-moment voxel carries the
 410 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$ — 36 un-
 411 knowns — tested with the same functions and coupled by cell-to-cell double
 412 integrals of the Green's tensor. For a laterally uniform layer at normal in-
 413 cidence it reduces *exactly* to Eq. (28) (`ContinuumLimit_Reduction3D.wl`).
 414 The lateral first moments are never excited: the lattice and the incident
 415 field are symmetric under reflection about every voxel's centre planes, and
 416 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 417 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-
 418 ishes on the reciprocal lattice and the plane sum of the double integrals

419 is the plate-to-plate integral in z alone. Two facts mark where genuinely
 420 three-dimensional structure begins. The lateral first moment's form fac-
 421 tor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the recipro-
 422 cal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the
 423 evanescent orders. And at oblique incidence the mean's own zeros shift,
 424 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h / p\pi + O(k_x^2 h^2)$, so the tiling identity holds
 425 only to $O(k_{\parallel} h)$.

426 *Oblique incidence..* The scheme is built spectrally. By Poisson summation
 427 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (30)$$

428 with L_a the lateral Legendre factors. The whole-space tensor decays as
 429 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each
 430 entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[i\alpha/2\gamma +$
 431 $i\beta \text{sgn}(z)/2] e^{i\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double
 432 averaging makes the lattice sum converge without Ewald splitting, to three
 433 digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20°
 434 and 40° (`ContinuumLimit_Oblique.wl`; Table 3). The mean-only voxel is
 435 second order and the first-moment voxel fourth order in both the reflected
 436 P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel} h)$
 437 and the evanescent coupling of the gradient channels do not cost an order.
 438 At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (28) to all printed digits.

439 *Incident S..* Nothing in the construction is specific to P. For a normally
 440 incident S wave the layer is the one-dimensional problem of Eq. (28) with

Table 3: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (30). At 40° the orders are again 2 and 4.

	mean only		mean and first moments	
n	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

441 $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta \mu$; the collocation and mean-only schemes
 442 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
 443 transmission, and the full 9×9 construction with an incident SH wave re-
 444 produces that one-dimensional result to all printed digits. At oblique inci-
 445 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
 are those of Table 4. The orders, and with $k = k_S$ the closed-form constants

Table 4: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

446

447 of Eqs. (26) and (29), are those of the incident P wave.

448 The spectral construction is implemented twice. The second implementa-
449 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
450 nition of the scheme: it takes the depth dependence of the transformed ker-
451 nel from an independent residue construction, obtains the local weight as a
452 numerical large- k_z limit of the inverted Christoffel matrix, evaluates every
453 double integral by quadrature and separates P from S by sampling the up-
454 going field at two depths. The raw reflected displacements of the two agree
455 to 9×10^{-16} in all 36 cases compared — incident P, SV and SH, normal
456 incidence and 20° , $n = 1, 2, 4$, with and without the first moments. Figure 1
457 collects the convergence.

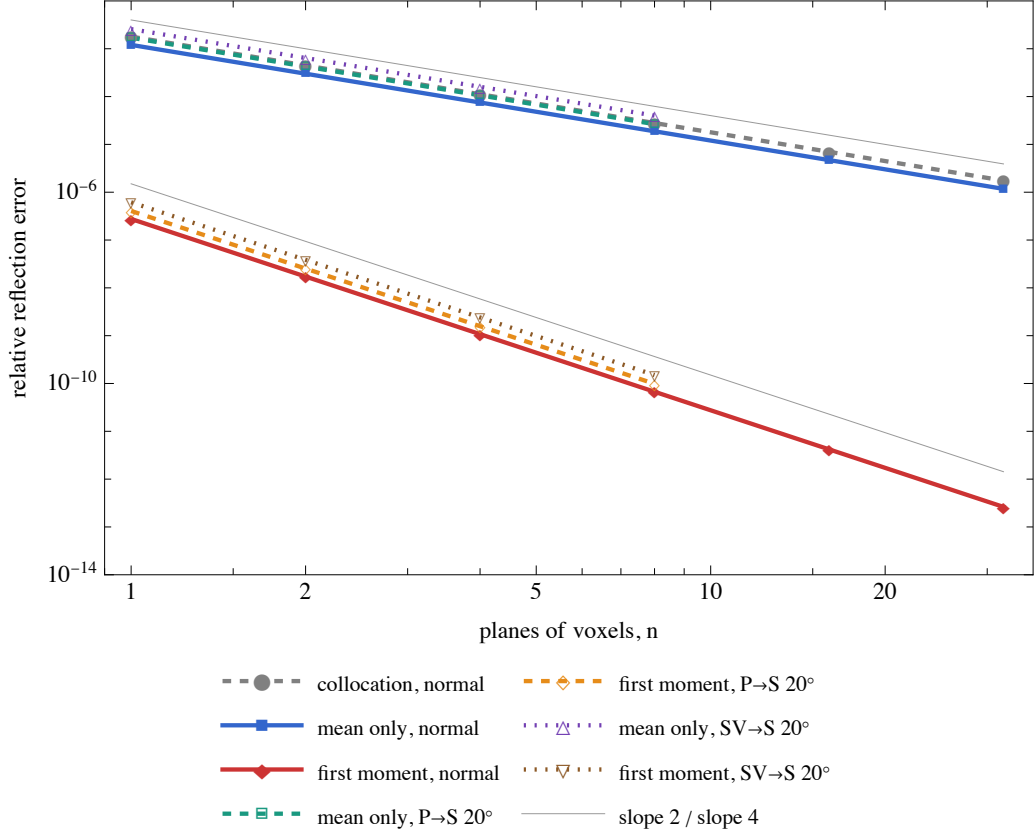


Figure 1: Relative reflection error of the discrete layer against the number of voxel planes n ($D = 2\text{ m}$, $\omega = 300\text{ rad s}^{-1}$, the contrast of §7). Filled: normal incidence, P reflection — collocation (§6), the mean-only voxel and the first-moment voxel of Eq. (28). Dashed, open squares and diamonds: P incident at 20° , the converted S wave, mean-only (green) and first-moment (orange) voxels of Eq. (30). Dotted, open triangles: SV incident at 20° , the reflected S wave, mean-only (purple) and first-moment (brown). Grey: slopes 2 and 4.

Alt text: A log-log plot of relative reflection error (vertical axis, 10^{-14} to 10^{-2}) against the number of voxel planes (horizontal axis, 1 to 32). The collocation and mean-only curves fall as straight lines of slope -2 from about 10^{-3} at one plane; the first-moment curves fall as straight lines of slope -4 from about 10^{-6} at one plane to about 10^{-13} at 32 planes. The oblique P-to-S and incident-SV curves lie alongside the normal-incidence curves of the same scheme, and grey guide lines show slopes 2 and 4.

458 9. A heterogeneous layer

459 A heterogeneous medium is not a layer of identical cells. The simplest
460 heterogeneous case with an exact answer is a stratified model built from layers
461 of constant properties, with real interfaces between them: the form of many
462 layered crustal velocity models. The last test takes such a model. A layer
463 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
464 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
465 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
466 stratification with physical contrasts. The voxels refine the model's layering,
467 with m planes of voxels in each model layer, so no voxel straddles a model
468 interface: the natural discretisation when the model's interfaces are known.
469 At $\omega = 1500 \text{ rad s}^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
470 reverberates between the model interfaces. The exact answer is the product
471 of the model layers' propagators. Its same-type reflection coefficients agree
472 with those of the reflection–transmission recursion of Kennett [25], computed
473 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
474 now changing from plane to plane; nothing else changes. The errors are in
475 Table 5.

476 *The interfaces cost nothing..* Every order is the one measured for the uniform
477 layer. At a model interface the strain of the true field jumps. A voxel basis
478 that is discontinuous from voxel to voxel represents that jump exactly when
479 the interface lies on a voxel boundary, so the interface adds no error of its
480 own. The only requirement is that voxel boundaries coincide with the model's
481 interfaces. A voxel that straddled an interface would carry a contrast that is
482 not uniform within it, a case not examined here.

Table 5: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

483 *What fourth order buys..* For a reflection error of 10^{-6} at normal incidence
484 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
485 layer. The first-moment voxel needs 8, one per model layer: the model's own
486 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
487 against 36), that is 3123 against 288 per vertical column of voxels, a factor
488 of eleven. In three dimensions, with the lateral grid refined as the vertical
489 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
490 unknowns by about 2×10^4 . The same model has been solved independently
491 in Python, with a contrast for each plane: the raw reflected displacements
492 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
493 at 20° , with and without the first moments. Lateral heterogeneity needs the
494 full three-dimensional solver; it is the first extension discussed below.

495 10. Discussion

496 *Where the error of a voxel scheme lives..* The results separate the two ap-
 497 proximations a voxel scheme makes. The *coupling* between voxels makes
 498 none: averaged over the source cell it is the continuum’s cell integral exactly,
 499 between planes by the form factor’s zeros and within a plane by the tiling
 500 identity, and the self term it omits is exactly the one the single site carries.
 501 Every error therefore comes from the *single site*, from the internal field it
 502 assumes. For a uniform field that error is the midpoint rule’s, known in
 503 closed form; for a linear field it is the next term of the same expansion. The
 504 converse is the practical warning. A kernel that is not the cell integral —
 505 the point Green’s tensor, or a cell average that does not match the single
 506 site’s closure — leaves a local term that does not vanish as the voxels shrink
 507 (§4): the evanescent lattice orders carry a static contribution of fixed size
 508 per voxel volume. A scheme built that way converges to the wrong limit,
 509 and refinement cannot reveal it, because the error is scale-invariant.

510 *What the single site is..* The moment hierarchy of §6 turns the uniform-field
 511 single site from a modelling choice into a derived object with a stated error.
 512 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
 513 only by the cubic split of Eq. (22). Its dynamic part departs from exact
 514 scattering by a sphere only through the real $(ka)^2$ term of Eq. (23); the
 515 radiation damping is exact. That term and the collocation error of Eq. (26)
 516 have the same origin, the variation of the field across the scatterer, and the
 517 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 518 field a voxel must represent is the smooth continuum field, not the field of
 519 an isolated cube, with its concentrations at the edges and corners.

520 *Relation to the discrete dipole approximation..* The same partition appears in
 521 the electromagnetic literature as the choice of polarisability. The Clausius–
 522 Mossotti relation, the lattice dispersion relation [6] and the integrated Green
 523 tensor [7] all adjust the single site so that the lattice reproduces the con-
 524 tinuum. For space-filling cubes the adjustment is neither needed nor a free
 525 parameter: the tiling identity fixes the local field exactly, the self term of the
 526 collocation closure is the one that makes it cancel, and what is left is a closed-
 527 form error rather than an empirical one. The second-order convergence re-
 528 ported for the DDA by Yurkin et al. [1] corresponds to the uniform-field
 529 voxel’s, and the opposite-signed constants of Eq. (26) explain why a polaris-
 530 ability correction can improve one observable only at the expense of another,
 531 as the package’s own form-factor corrections do in reflection and transmis-
 532 sion. Going beyond second order requires a richer internal field, which is the
 533 point of higher-order bases for volume integral equations; the first-moment
 534 voxel is the lowest such basis, and in its spectral form the Galerkin double
 535 average makes the lattice sum converge without Ewald splitting.

536 *The cost of fourth order..* A first-moment voxel carries four times as many
 537 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 538 forms show what that buys. To reach an error ε in reflection, the uniform
 539 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 540 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 541 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 542 number of unknowns by about 5×10^4 (these are the Born-order constants;
 543 at the contrast studied the first-moment voxel’s measured errors lie within
 544 5% of them, while the uniform voxel’s transmission constant grows with the

contrast, so the comparison is if anything conservative). The stratified model of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-moment voxels for the same accuracy. The gain is largest precisely where it matters: when the model varies on scales that are coarse compared with the accuracy the wavefield must have.

Limitations and outlook. The results concern laterally uniform media — a homogeneous layer and a randomly stratified one — for which the discrete system is exactly a chain of planes; a laterally heterogeneous medium needs the first-moment voxel in a full three-dimensional solver, where the gradient channels couple through the evanescent lattice orders (§8) and the tiling identity holds only to $O(k_{\parallel}h)$. The numerical evidence is at moderate, physically typical contrasts (up to 15% in density and moduli) and at $k_S h \leq 0.3$, the range in which the single-site T -matrix is validated; the closed forms are first order in the contrast, and the departures from them at the full contrast are of second order. What is established is the *order* of convergence and its leading constant, not the behaviour at strong contrast, where the single site’s own closure may limit the accuracy before the lattice coupling does. The single site’s dynamic error is in closed form for a sphere only; for a cube, for which no exact solution exists, it is known through the layer (§§7–8). Two extensions follow directly. The first is the three-dimensional solver itself. The second is a stratified *background* — a layered reference medium, as distinct from the stratified contrast of §9: the lattice coupling then carries the reverberation of the layering, which is not translation invariant in depth, and whether a tiling identity holds for it is the natural next question.

569 11. Conclusions

570 A lattice of cubic voxels reproduces a continuous elastic medium in a pre-
571 cise sense. Because cubes tile the plane, the source-cell-averaged coupling be-
572 tween voxels is the continuum’s cell integral exactly, and at normal incidence
573 the lattice kernel is available in closed form with no lattice sum. The single
574 site is derived rather than assumed. The moments of the Green’s tensor over
575 the cube, taken as distributions, give a closed hierarchy whose first-gradient
576 truncation is the uniform-strain closure. That closure is exact for a sphere
577 in the static limit, and it departs from exact scattering by a sphere only by a
578 real $(ka)^2$ term, with the radiation damping exact. For a cube it differs from
579 the sphere only by a cubic anisotropy of the shear response. The collocation
580 closure carries the same self term as the lattice kernel, which cancels, so the
581 discrete system is a collocation of the continuum equation. It converges at
582 second order, with a leading error that is a midpoint-rule error of the internal
583 field, known in closed form for every discretisation, and opposite in sign in
584 reflection and transmission. A voxel that also carries the first moment of
585 its internal field converges at fourth order — at normal incidence, where the
586 three-dimensional scheme reduces exactly to one dimension, and at oblique
587 incidence — with a leading error, $-(kd)^4/720$, that no longer depends on
588 the direction of observation. The orders survive the interfaces of a randomly
589 stratified model, provided voxel boundaries coincide with its interfaces, and
590 there the first-moment voxel reaches an accuracy of 10^{-6} with one voxel per
591 model layer, where the uniform-field voxel needs about forty. The results,
592 and the evidence for each, are collected in Table 6.

Table 6: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (9)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (10)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (17)	sum rule (18) exact; M, Q equal an independent archive, 81/81
Channels (19)–(21)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (22)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube’s bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (23)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (25)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (26)	$n = 1\text{--}16$, to 10^{-4} of the error
First-moment voxel, Eq. (28)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (30)	order 4, P and S, 20° and 40°
Incident SV and SH, all schemes	orders 2 and 4; constants with k_S to 4×10^{-4}

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595 the Research School of Earth Sciences of the Australian National University,
596 Canberra [9].

597 **Declaration of competing interest**

598 The author declares no known competing financial interests or personal
599 relationships that could have appeared to influence the work reported in this
600 paper.

601 **Data availability**

602 The Mathematica notebooks and Python scripts that derive and check ev-
603 ery result, and the package they test, are archived at [https://doi.org/10.](https://doi.org/10.5281/zenodo.23003242)
604 5281/zenodo.23003242 and developed at [https://github.com/tmnestor/](https://github.com/tmnestor/MultipleScatteringCalculations)
605 MultipleScatteringCalculations.

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